Week – 1 Finding Time Complexity of Algorithms

Q1. Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
    int s =1;
    while(s <= n)
    {
        i++;
        s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and printf() statements. count variable

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>

int count = 0;

void function(int n){
    int i=1;
    count++;
    int s=1;
    count++;
    while(s<=n){
        count++;
        count++;
        i++;
        s+=i;
        count++;
    }
    count++;
    printf("%d",count);
}

int main(){
    int n;
    scanf("%d",&n);
    function(n);
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Q2. Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and printf() statements. count variable

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>

int count = 0;
void func(int n){
    if(n==1){
        count++;
        printf("*");
    } else {
        count++;
        for(int i=1; i<=n; i++){
            count++;
            for(int j=1; j<=n; j++){
                count++;
                count++;
                count++;
                break;
            }
            count++;
        }
        count++;
    }
    printf("%d",count);
}

int main(){
    int n;
    scanf("%d",&n);
    func(n);
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Q3. Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num;++i)
    {
     if (num % i== 0)
         {
           printf("%d ", i);
         }
     }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include<stdio.h>

int count=0;
void Factor(int num){
    for(int i=1; i<=num; ++i){
        count++;
        if(num%i==0){
            count++;
            //printf("%d ",i);
        }
        count++;
    }
    count++;
    printf("%d",count);
}

int main(){
    int n;
    scanf("%d",&n);
    Factor(n);
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Q4. Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and printf() statements. count variable

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include<stdio.h>

int count = 0;
void function(int n){
    int c = 0;
    count++;
    for(int i=n/2; i<n; i++){
        count++;
        for(int j=1; j<n; j=2*j){
            count++;
            for(int k=1; k<n; k = k*2){
                count++;
                c++;
                count++;
            }
            count++;
        }
        count++;
    }
    count++;
    printf("%d",count);
}

int main(){
    int n;
    scanf("%d",&n);
    function(n);
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Q5. Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and printf() statements. count variable

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include<stdio.h>

int count = 0;
void reverse(int n){
    int rev = 0, reminder;
    count++;
    while(n!=0){
        count++;
        reminder = n%10;
        count++;
        rev = rev * 10 + reminder;
        count++;
        n/=10;
        count++;
    }
    count++;
    count++;
    printf("%d",count);
}

int main(){
    int n;
    scanf("%d",&n);
    reverse(n);
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔